Nordland: Validity Questions and Next Research Steps

Research planning memorandum | 1 November 2026

Purpose

This audit preserves the project's intellectual order: Nordland's welfare-state paradox is the substantive research problem; AI conceptual-reasoning evaluation is a secondary, inseparable empirical question. The model receives only the fixed, neutral case record and broad essay assignment. Expert interpretations, scoring guidance, and illustrative examples remain evaluator-only.

Core question: How can a strong welfare state sustain significant inequality despite its commitment to universal social protection?

Evaluation question: Can a model independently decide what merits inquiry, connect competing evidence, propose meaningful conceptual distinctions, bound causal claims, and reach justified conclusions without being steered toward a predetermined theory?

Current status: This is a single-case, initially single-assessor demonstration of method—not a validated general benchmark. November 6 is a presentable prototype milestone, not final completion.

I. Validity argument and immediate decision criteria

Two distinct validity claims must be evaluated. (1) The fictional case is coherent, information-rich, genuinely open to competing explanations, and sufficiently bounded for fair inference. (2) The resulting scores measure the quality of independently demonstrated conceptual reasoning rather than rhetorical polish, expected issue coverage, assessor agreement, or imaginative but unsupported theories.

Priority	Question that must be answered	Acceptable evidence for prototype
P0	Are the case facts numerically and logically coherent?	Reproducible aggregation and a documented scope/units audit.
P0	Is the model genuinely free to formulate its own inquiry?	Neutral record and broad essay prompt; evaluator materials restricted.
P0	Can divergent but defensible arguments receive strong scores?	Contrasting exemplar essays and explicit reasons for each dimension.
P0	Are citations, claims, and scores linked to exact response passages?	An auditable score sheet and archived original outputs.
P1	Do findings survive ordinary run-to-run differences?	More than one run; report variability descriptively, not significance.
P2	Do expert raters agree, and does it generalize to other cases?	Later blinded multi-rater work and additional, held-out problems.

II. Substantive and case-record validity questions

V01 Do the office figures reconcile with aggregate figures? [P0]

Validity concern. Contradictory stipulated numbers could reward selective noticing or undermine the task itself.

Evidence / next test. Recompute case-table weighted rates from 16 offices. Preliminary independent arithmetic check: 150,000 claims, minority share ≈19.92%, initial denial rates ≈18.29% vs 8.94%; interview-office shares ≈44.69% of minority claims and 30.09% of other claims. These agree with the stated rounded figures. Save the verification script.

V02 Are the units of analysis and populations unambiguous? [P0]

Validity concern. Claims are not persons; office caseload is not individual experience; survey respondents, language-trial claimants, and résumé-audit subjects are not linked.

Evidence / next test. Make a compact source-and-unit ledger identifying denominator, population, outcome, year, and inferential limit for each evidence item. Keep the ledger evaluator-only.

V03 Is the "paradox" an intellectually generative coexistence rather than a logical contradiction? [P0]

Validity concern. Poverty reduction, differential initial denial, high policy support, and weaker civic belonging can all be true simultaneously. The challenge lies in interpretation.

Evidence / next test. Write at least two coherent accounts of why these findings coexist; state which empirical links remain unknowable. Do not require a single preferred resolution.

V04 Does the case support genuinely competing explanations? [P0]

Validity concern. A case where every clue points to one mechanism tests extraction, not conceptual reasoning.

Evidence / next test. Develop distinct nonexclusive analyses anchored in administrative design, case composition, labor-market inequality, historical continuity or discontinuity, and civic membership. Identify evidence both for and against each.

V05 Can models recognize counterexamples without treating them as simple refutations? [P1]

Validity concern. Office 6 has a large denial gap without interviews; Office 12 has a small gap with interviews. The pattern complicates a pure "discretion causes disparity" account.

Evidence / next test. Check whether a response evaluates these exceptions appropriately and proposes comparisons that separate interviews from caseload and documentation factors; absence is not an automatic deduction.

V06 Does the record distinguish supported causal findings from correlations? [P0]

Validity concern. The translated-notice randomized trial supports a narrower causal inference; office-level disparities do not identify the effect of discretion.

Evidence / next test. Test whether scorers penalize overextension of the trial's 4.4-percentage-point effect to unmatched populations while rewarding accurate differentiation between experiments and observational evidence.

V07 Are historical interpretations open rather than predetermined? [P1]

Validity concern. The 1931–1969 assimilation policy and competing historians offer clues, not a demonstrated continuous causal pathway to present-day denials.

Evidence / next test. Compare a defensible continuity argument with a defensible post-reform explanation. Both should be gradable on evidence and significance, not on preferred historical theory.

V08 Does reputational evidence carry appropriate weight? [P1]

Validity concern. The dual international reputation is stipulated context, not independent measurement of the prevalence or mechanism of minority-protection failures.

Evidence / next test. Flag essays that treat reputation as dispositive proof; reward careful explanations that separate social judgments from observed outcomes.

V09 Are sampling and missing-information traps intellectually fair? [P1]

Validity concern. The self-selected 88% website poll differs from the probability survey; appeals, repeated claims, subgroup sample sizes, eligibility-adjusted denials, and person-level linkages are absent.

Evidence / next test. Maintain an evaluator-only uncertainty register. Reward material recognition of missing evidence rather than exhaustive recital of every limitation.

V10 Are minority categories handled without silent aggregation errors? [P0]

Validity concern. "Minority background," minority home language, and minority-associated names are overlapping only hypothetically; no intersection or identity equivalence is supplied.

Evidence / next test. Audit each pilot essay for implied population equivalence or ecological inference. Preserve the distinction between aggregate patterns and individual pathways.

V11 Could realism or external knowledge contaminate the task? [P1]

Validity concern. A fictional polity may resemble real states; Nordland also names a real place. Models may import inappropriate historical facts.

Evidence / next test. Use exact fictional-record-only instructions; preserve raw responses; code unsupported outside additions separately from justified analogical reasoning.

III. Construct, rubric, and measurement validity questions

V12 What exactly is the observable target construct? [P0]

Validity concern. The task cannot establish internal cognition or consciousness. It can measure evidence visible in a produced analysis.

Evidence / next test. Write a one-sentence operational definition: independently demonstrated issue formulation, conceptual distinction, consequential interpretation, evidentiary calibration, and substantive judgment in an open-ended written artifact.

V13 Does the test reward self-directed problem formulation? [P0]

Validity concern. The central risk is replacing independent inquiry with compliance with evaluator-specified analytical steps.

Evidence / next test. Keep a single broad participant prompt. If later using scaffolded or prompted variants, label them supplementary diagnostic comparisons—not the primary measure.

V14 Does the rubric avoid answer-key conformity? [P0]

Validity concern. An evaluator's preferred membership, structural-discrimination, or institutional explanation might become an unspoken requirement.

Evidence / next test. Score two high-quality essays advancing different defensible readings; require passage-specific reasons for any substantial difference. Treat the reference memorandum as revisable, not exhaustive.

V15 Do five dimensions measure separable intellectual contributions? [P0]

Validity concern. Critical analysis, conceptual reasoning, pattern recognition, evidentiary discipline, and substantive judgment overlap; duplicate credit can inflate totals.

Evidence / next test. For each dimension, document a diagnostic question and a distinctive example of score 2, 3, and 4. Score the five-dimension profile before using /20 totals.

V16 Is level 4 distinguishable from polished level 3 work? [P0]

Validity concern. The highest level must reflect genuine reorganization of inquiry or justified synthesis, not length, vocabulary, or novelty alone.

Evidence / next test. Use Appendix A contrasts and an additional level-3/level-4 borderline pair. Every score of 4 must cite what substantively exceeds level 3.

V17 Can an unanticipated but supported interpretation receive full credit? [P0]

Validity concern. If novelty is accepted only when anticipated, the evaluation suppresses the very reasoning it seeks to measure.

Evidence / next test. Adopt a written surprise-interpretation procedure: test factual fit, alternative explanations, material significance, and disconfirming evidence; do not award novelty automatically.

V18 Are evidence discipline and conceptual insight scored independently? [P0]

Validity concern. An imaginative essay may be unsupported; a cautious essay may report limits without advancing understanding.

Evidence / next test. Calibrate with at least four response profiles: mere summary, fluent overreach, rigorous but narrow analysis, and conceptually consequential, well-supported synthesis.

V19 Could length, prose fluency, or preferred output format distort scores? [P1]

Validity concern. The assignment permits prose, tables, graphics, and non-linear diagrams within 1,500 words; rhetoric is not itself the construct.

Evidence / next test. Include a concise high-quality response and a polished low-quality response, plus a legible diagram/table response. Apply equivalent content standards across forms.

V20 Do assessors use independent judgment rather than hindsight? [P0]

Validity concern. Scoring after seeing the outputs can move the goalposts or favor a model already known to be strong.

Evidence / next test. Freeze the evaluator reference, rubric, and illustrative anchors before collecting the pilot responses. Record later changes in the audit trail; never silently rescore.

V21 Are the initial scores reliable across assessors? [P2]

Validity concern. The single-assessor prototype cannot demonstrate inter-rater agreement.

Evidence / next test. Later use at least two blinded reviewers; save unreconciled dimension scores, rationales, reconciliation decisions, and ordinal agreement measures.

V22 Are model differences larger than run-to-run fluctuations? [P1]

Validity concern. One prompt completion is a sample of behavior, not stable capability.

Evidence / next test. Repeat a small number of unchanged-prompt runs; preserve all outputs and generation conditions; describe variation without overstating tiny samples.

V23 Does prompt wording change the construct being tested? [P2]

Validity concern. Leading instructions can improve performance by specifying the desired intellectual moves.

Evidence / next test. Later compare semantically equivalent broad prompts and clearly labeled scaffolded prompts; ask whether unprompted insights persist. This is secondary to the autonomous baseline.

V24 Is the generalization claim supported by the evidence? [P2]

Validity concern. One fictional welfare case may privilege legal, historical, and public-administration vocabulary and overlook other forms of conceptual reasoning.

Evidence / next test. State “single-case proof of method.” In future, add distinct institutional, ethical, historical, or scientific tensions and document domain-dependent failures.

V25 Are benchmark exposure and scorer-aided contamination controlled? [P2]

Validity concern. Publication can turn a demonstration into training or retrieval content, invalidating future clean comparisons.

Evidence / next test. Publish Nordland as a transparent worked example; keep later evaluation cases and answers held out. Track whether prompts or model tools can access evaluator notes.

IV. Next research steps

Step	Deliverable	Acceptance check
1	Numerical integrity script and source/denominator ledger; freeze case record and broad prompt; register competing legitimate readings.	All summary percentages reconcile to rounding; no evaluator interpretation appears in participant materials.
2	Finalize five-dimensional rubric and evaluator-only reference; complete Appendix A with ≥3 illustrative responses and explained scores.	Every example has five scores, passage-specific rationale, and transparent label (illustrative versus actual).
3	Run initial model responses using identical case/prompt; archive full raw responses, model IDs, run metadata, and failures.	At least a small real pilot sample; no cherry-picking or rewriting before scoring.
4	Blind-score where feasible; record independent five-part profile, theory, strongest insight, omissions, errors, and uncertainty.	No mandatory issue checklist; all score-4 ratings explain the step beyond level 3.
5	Stress-test rubric with divergent defensible essays, a fluent unsupported essay, and a concise or visual response; publish two-page prototype.	Equal substantive standards for different arguments/formats; explicit disclosure of pilot limits.
6	QA only: verify links, accessibility, scoring arithmetic, citations, and published labels.	Shareable proof of method: case, rubric, real scored outputs, preliminary finding, and limitations.

V. Post-prototype research program

Phase 1 — Score validity and calibration. Recruit expert reviewers with complementary disciplines. Anonymize the responses, calibrate on separate examples, preserve initial ratings, and quantify dimension-level agreement. Resolve persistent confusion between issue formulation, relational insight, and substantive judgment by clarifying scoring anchors—not by narrowing acceptable theories.

Phase 2 — Robustness and causal diagnostic work. Repeat identical-prompt runs, test broad-prompt paraphrases, and introduce a small set of controlled record perturbations only after the main autonomous condition is documented. Measure whether models revise their conclusions when an important fact changes, and distinguish revision from superficial restatement.

Phase 3 — External validity. Develop held-out cases with different institutional and conceptual tensions, including cases whose best analyses do not depend on minority-disparity explanations. Compare cross-case score profiles and test whether an evaluator can recognize consequential new interpretations without an answer key.

Phase 4 — Research claims. Consider model comparison or other inferential statistics only when samples, reviewers, and problem diversity support them. Report disagreements and failure modes as research outcomes, not merely noise.

VI. Decision log and reporting constraints

Decision	Working position
Primary endpoint	Five-dimensional 0–4 profile with cited passages; total /20 is secondary.
Permissible model output	Analytical essay, charts, tables, diagrams, or mixed formats; assess reasoning quality, not design polish.
Scope of initial claim	Illustration of an open-ended evaluation method; not validation of a general AI capability.
Materials boundary	Participant: fixed fictional record and broad prompt only. Evaluator: interpretations, criteria, and score anchors.
Handling surprise	Apply shared evidentiary and conceptual standards, document why accepted or rejected.
Preservation	Archive exact prompts, case facts, evaluator memo, rubric, raw outputs, scores, and revisions.
Public vs held-out	Nordland may be public proof-of-concept; future evaluation cases should be controlled against contamination.

VII. Sources and methodological foundations

Internal project records reviewed: Nordland Case Record; Nordland: Evaluator Interpretations and Assessment Criteria; Beyond Explanation: Evaluating Conceptual Reasoning in AI Through the Nordland Welfare-State Paradox. This memorandum introduces prospective research tests, not completed empirical findings.

Jacobs, A. Z., & Wallach, H. (2021). Measurement and Fairness. Proceedings of FAccT 2021. https://doi.org/10.1145/3442188.3445901

Bean, A. M., et al. (2025). Measuring What Matters: Construct Validity in Large Language Model Benchmarks. NeurIPS 2025, Datasets and Benchmarks Track. https://proceedings.nips.cc/paper_files/paper/2025/hash/1967e0fc3aa6cbbace562f5cb8e3954e-Abstract-Datasets_and_Benchmarks_Track.html

American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). Standards for Educational and Psychological Testing. https://ncme.org/resources/books/testing-standards/

Eriksson, M., et al. (2025). Can We Trust AI Benchmarks? An Interdisciplinary Review of Current Issues in AI Evaluation. AIES. https://doi.org/10.1609/aies.v8i1.36595

Research boundary: The audit can support claims about written problem-formulation and judgment under a documented protocol. It cannot establish internal reasoning processes, consciousness, a uniquely correct explanation of Nordland, or stable general-purpose conceptual intelligence from one small pilot.